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Abstract

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1 Introduction

2 Theory / Background

We begin by establishing the theoretical kinematic dynamics governing the swarm of birds, and then using these dynamics to infer the flow of information through the system. Analysis of several videos of swarming shows that a full swarm may be comprised of several sub-swarms. For the purposes of this paper, we will denote each of these sub-swarms as a *component* [You25].

Definition 2.1. A **component** is defined as the tuple (C, r), a set of birds, C, such that any bird in the component may be reached from any other bird in the component via a sequence of birds, each within radius r of the next.

Assumption 2.2. *All swarms are composed of a finite number of components.*

> Rationale: *There exists a finite number of birds in the universe (hopefully) and thus even in the worst case scenario where each bird is its own component, there will still be a finite number of components.*

Assumption 2.3. *The behavior of birds within a component (C, r) is independent of the behavior of birds outside the component.*

> Rationale: *Information and therefore influence may only be transmitted bird to bird within a limited radius, for convienience assume r is chosen to be less than or equal to this radius.*

Under assumptions 2.1 and 2.2 it suffices to analyze the behavior of a single component. So for now consider a single component (C, r). Let $N = |C| < \infty$ be the number of birds in the component.

Definition 2.4. A **state** at time t is a vector

$$\Psi(t) = (\theta_1(t), \theta_2(t), \ldots, \theta_N(t), X_1(t), X_2(t), \ldots, X_N(t)) \in \mathbb{R}^{6N},$$

where $\theta_i(t), X_i(t) \in \mathbb{R}^3$ represent the heading and position of bird i in space at time t (heading taken to be normalized).

Let Ψ_0 be the initial state of the system at time $t = 0$. The evolution of the system can then be modeled by the following initial value problem

$$\begin{cases} \dfrac{\mathrm{d}\Psi(t)}{\mathrm{d}t} = F(\Psi(t)), \\ \Psi(0) = \Psi_0 \end{cases} \tag{2.1}$$

where $F : \mathbb{R}^{6N} \to \mathbb{R}^{6N}$ is a function that encodes interactions between birds in the system.

Assumption 2.5. *F is continuous and differentiable in Ψ on a neighborhood of Ψ_0.*

> Rationale: *Being a physical system, between macroscopic objects (birds) we should expect the interactions to be continuous and at least C^1 smooth.*

Assumption 2.6. *Under small time scales the relative separations, $\|X_i - X_j\|$, of birds in the swarm can be considered constant.*

> Rationale: *Studies have determined that topological properties (i.e. closest neighbors not metric distance of closest neighbors) matter more in bird flocking behavior than metric distance, so for small time scales we can assume that distances between birds remain constant without affecting end behavior of system [Bal+08]*

Looking at assumption 2.5 let us now consider its implications. First we can define the separation $N \times N$ matrix $K(t)$, constructed as $K_{i,j}(t) = \|X_i(t) - X_j(t)\|$:

$$K(t) = \begin{bmatrix} \|X_1(t) - X_1(t)\| & \|X_1(t) - X_2(t)\| & \cdots & \|X_1(t) - X_N(t)\| \\ \|X_2(t) - X_1(t)\| & \|X_2(t) - X_2(t)\| & \cdots & \|X_2(t) - X_N(t)\| \\ \vdots & \vdots & \ddots & \vdots \\ \|X_N(t) - X_1(t)\| & \|X_N(t) - X_2(t)\| & \cdots & \|X_N(t) - X_N(t)\| \end{bmatrix} \tag{2.2}$$

Under a small time scale $t \in (t_0, t_0 + \delta)$, we can consider $K(t)$ to be constant matrix by assumption 2.5, which we will denote K. The final simplification we make is to furhter restrict positions $X_i(t)$ to be in a lattice structure, to simplfiy analysis and neiborhood computations.

Assumption 2.7. *Under a small time scale, the positions $X_i(t)$ of birds in the component are fixed on a lattice structure in $\mathbb{R}^3$.*

<u>Rationale:</u> From assumption 2.5 we know that under a small time scale the relative separations of birds remain constant, now also referring to [Bal+08] we know that birds focus more on topological neighbors rather than metric neighbors, so since for this model we are more concerned about turning and heading information rather than exact positions we can assume a lattice structure.

Now for a small time scale equation (2.1) can be rewritten as a system of first oder ODEs

$$\begin{cases} \dfrac{\mathrm{d}\theta_i(t)}{\mathrm{d}t} = f_i(\theta_1(t), \theta_2(t), \ldots, \theta_N(t); K), & i = 1, 2, \ldots, N, \\ \theta_i(0) = \theta_{i,0}, & i = 1, 2, \ldots, N, \end{cases} \tag{2.3}$$

Theorem 2.8 (Existence and Uniqueness of Solutions to (2.3))**.** *Let F be as defined in (2.1), then there exists a unique local solution corresponding to the IVP (2.3)*

Proof. From assumption 2.5 we know that F is differentiable and therefore by MVT locally lipschitz in Ψ on a neighborhood of Ψ_0. Now it follows that for each f_i in (2.3), f_i is also locally lipschitz in θ_j on a neighborhood of $\theta_{j,0}$. therefore by Picard–Lindelöf theorem there exist unique solutions (2.3) for small time scales. □

The remainder of this paper will focus on analyzing the solutions to this system of equations, with a second order variation of this system being considered in section [TBD].

3 Swarm on a Sphere Analysis

As discussed in the introduction the two main methods we will use to analyze the first-order system (2.3), are the Swarm on a Sphere (SOS) model and the Generalize Ising Model. First we will examine this system through the SOS approach. To build up to SOS we first introduce the Kuramoto model. The Kuramoto model is a model in mathematical physics to describe a system of synchronizing oscillators. that take the from

$$\begin{cases} \frac{d\psi_i}{dt} = \omega_i + \frac{1}{N}\sum_{j=1}^{N} R_{ij} sin(\psi_j - \psi_i) \\ \psi_i(0) = \psi_{i,0} \end{cases} \tag{3.1}$$

Where ψ_i represents the phase of the ith oscillator, ω_i its natural frequency, and R_{ij} the coupling strength between oscillators i and j.
While I would have loved to use Kuramoto to model the heading in (2.3), the main limitation is that $\psi_i(t) \in \mathbb{R}^2$, while accurately modeling (2.3) requires $\theta_i(t) \in \mathbb{R}^3$ (again assumed normalized).
The generalization of this model to $d - dim$ is the the Swarm on a Sphere (SOS) model [Olf06], which models coupled oscillators on $\mathbb{R}^d$. Where in our context $d = 2$ and the oscillators represent the heading of each bird in the swarm. SOS is given by the follwing

$$\frac{dx_i}{dt} = \sum_{j=1}^{N} R_{ij}(x_j - \langle x_j, x_i \rangle x_i) \tag{3.2}$$

Where $x_i \in \mathbb{R}^d$ is the unit vector of bird i in the swarm, and R_{ij} the interaction strength between birds i and j. We get that $\theta_i = x_i$. and thus f_i is given by the RHS of (3.2). Before analyzing (3.2) analytically, we first numerically experiment to determine key parameters that affect the behavior of the system, then we analyze the system to determine stability and long term behavior of solutions.

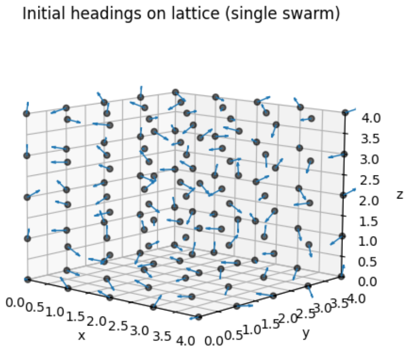

Figure 1: Visualization of Initial Heading Allginents on Lattice

3.1 Computational Experiments

3.1.1 Implementation

The goal of the computational experimetns was to numerically determine 2 key parameters that affect the behavior, namely the intial allginment and the interaction matrix R.

Definition 3.1. The **initial alignment** of a component (C, r) is defined as

$$A = \frac{1}{N} \left\| \sum_{i=1}^{N} \theta_{i,0} \right\| \in [0, 1].$$

Where $A = 1$ represents perfect initial allignment (all birds facing the same direction) and $A = 0$ represents initial disallignment (birds in random directions).

To accomplish this I wrote a python simulation that numerically solves (3.2) on a 3D lattice using scip's IVP solver. The simulation works by approximating a component (C, r) on a $X \times Y \times Z$ 3D lattice with edge length r between lattice points, and $N = X \times Y \times Z$. Each vertex represents a bird with an associated heading (unit vector in $\mathbb{R}^3$).

To intialzie the simulation takes in the size of the flock (X, Y, Z), an initial alignment A which is what determiens the distribution of initial headings of the birds, and an intteraction matrix R. Since from [Bal+08] we know that interactions are based off of topological neighbors, the interaction matrix R is constructed based off of **order**, and **weight**.

Definition 3.2. The **order** of the interaction matrix R on a lattice L, is defined as the maximum lattice "hops" two birds may be and still interact. (i.e. order 1 means birds only interact with direct lattice neighbors, order 2 neighbors of neighbors and so on).

Definition 3.3. The **weight** of the interaction matrix R on a lattice L, is defined as the strength of interaction between birds that are at the maximum order of interaction.

As it is obvious that if all else is kept equal, a higher order will lead to quicker swarm alignment, to accurately determine the appropriate order I give every bird a constant "attention", At such that

$$At = \sum_{j=1}^{N} R_{ij}, \quad \forall i = 1, 2, \ldots, N. \tag{3.3}$$

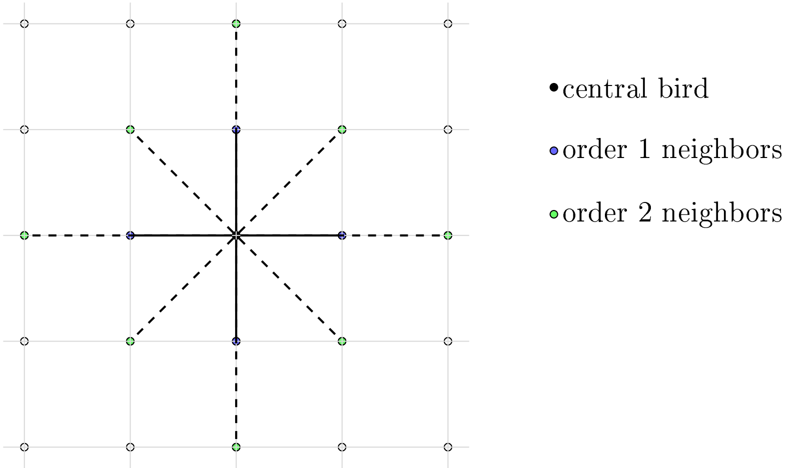

Figure 2: Order 1 and 2 interaction on a 2D lattice (AI Disclaimer: I used chatGPT to generate this tikz drawing)

This means that as order increases, the weight must decrease simulating that it is harder for brids to pay attention to more birds at any given time. I finally ran several simulation over a sweep of initial alignments, orders, and weights, over a variety of flock sizes to determine which values created realistic flocking behavior.

3.1.2 Results

4 Methods

5 Results

6 Discussion

7 Conclusion

Acknowledgments

A Supplementary material

References

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